

4 CONVERGENCE ANALYSIS

We analyze the convergence of Adam using the online learning framework proposed in (Zinkevich, 2003). Given an arbitrary, unknown sequence of convex cost functions $f_1(\theta), f_2(\theta), \dots, f_T(\theta)$. At each time t , our goal is to predict the parameter θ_t and evaluate it on a previously unknown cost function f_t . Since the nature of the sequence is unknown in advance, we evaluate our algorithm using the regret, that is the sum of all the previous difference between the online prediction $f_t(\theta_t)$ and the best fixed point parameter $f_t(\theta^*)$ from a feasible set $\mathcal{X}$ for all the previous steps. Concretely, the regret is defined as:

$$R(T) = \sum_{t=1}^T [f_t(\theta_t) - f_t(\theta^*)] \quad (5)$$

where $\theta^* = \arg \min_{\theta \in \mathcal{X}} \sum_{t=1}^T f_t(\theta)$. We show Adam has $O(\sqrt{T})$ regret bound and a proof is given in the appendix. Our result is comparable to the best known bound for this general convex online learning problem. We also use some definitions simplify our notation, where $g_t \triangleq \nabla f_t(\theta_t)$ and $g_{t,i}$ as the i^{th} element. We define $g_{1:t,i} \in \mathbb{R}^t$ as a vector that contains the i^{th} dimension of the gradients over all iterations till t , $g_{1:t,i} = [g_{1,i}, g_{2,i}, \dots, g_{t,i}]$. Also, we define $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$. Our following theorem holds when the learning rate α_t is decaying at a rate of $t^{-\frac{1}{2}}$ and first moment running average coefficient $\beta_{1,t}$ decay exponentially with λ , that is typically close to 1, e.g. $1 - 10^{-8}$.

Theorem 4.1. *Assume that the function f_t has bounded gradients, $\|\nabla f_t(\theta)\|_2 \leq G$, $\|\nabla f_t(\theta)\|_\infty \leq G_\infty$ for all $\theta \in \mathbb{R}^d$ and distance between any θ_t generated by Adam is bounded, $\|\theta_n - \theta_m\|_2 \leq D$, $\|\theta_m - \theta_n\|_\infty \leq D_\infty$ for any $m, n \in \{1, \dots, T\}$, and $\beta_1, \beta_2 \in [0, 1)$ satisfy $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$. Let $\alpha_t = \frac{\alpha}{\sqrt{t}}$ and $\beta_{1,t} = \beta_1 \lambda^{t-1}$, $\lambda \in (0, 1)$. Adam achieves the following guarantee, for all $T \geq 1$.*

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^d \sqrt{T \hat{v}_{T,i}} + \frac{\alpha(1+\beta_1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \sum_{i=1}^d \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha(1-\beta_1)(1-\lambda)^2}$$

Our Theorem 4.1 implies when the data features are sparse and bounded gradients, the summation term can be much smaller than its upper bound $\sum_{i=1}^d \|g_{1:T,i}\|_2 \ll dG_\infty \sqrt{T}$ and $\sum_{i=1}^d \sqrt{T \hat{v}_{T,i}} \ll dG_\infty \sqrt{T}$, in particular if the class of function and data features are in the form of section 1.2 in (Duchi et al., 2011). Their results for the expected value $\mathbb{E}[\sum_{i=1}^d \|g_{1:T,i}\|_2]$ also apply to Adam. In particular, the adaptive method, such as Adam and Adagrad, can achieve $O(\log d \sqrt{T})$, an improvement over $O(\sqrt{dT})$ for the non-adaptive method. Decaying $\beta_{1,t}$ towards zero is important in our theoretical analysis and also matches previous empirical findings, e.g. (Sutskever et al., 2013) suggests reducing the momentum coefficient in the end of training can improve convergence.

Finally, we can show the average regret of Adam converges,

Corollary 4.2. *Assume that the function f_t has bounded gradients, $\|\nabla f_t(\theta)\|_2 \leq G$, $\|\nabla f_t(\theta)\|_\infty \leq G_\infty$ for all $\theta \in \mathbb{R}^d$ and distance between any θ_t generated by Adam is bounded, $\|\theta_n - \theta_m\|_2 \leq D$, $\|\theta_m - \theta_n\|_\infty \leq D_\infty$ for any $m, n \in \{1, \dots, T\}$. Adam achieves the following guarantee, for all $T \geq 1$.*

$$\frac{R(T)}{T} = O\left(\frac{1}{\sqrt{T}}\right)$$

This result can be obtained by using Theorem 4.1 and $\sum_{i=1}^d \|g_{1:T,i}\|_2 \leq dG_\infty \sqrt{T}$. Thus, $\lim_{T \rightarrow \infty} \frac{R(T)}{T} = 0$.

5 RELATED WORK

Optimization methods bearing a direct relation to Adam are RMSProp (Tieleman & Hinton, 2012; Graves, 2013) and AdaGrad (Duchi et al., 2011); these relationships are discussed below. Other stochastic optimization methods include vSGD (Schaul et al., 2012), AdaDelta (Zeiler, 2012) and the natural Newton method from Roux & Fitzgibbon (2010), all setting stepsizes by estimating curvature

$\|\theta_m - \theta_n\|_\infty \leq D_\infty$ for any $m, n \in \{1, \dots, T\}$, and $\beta_1, \beta_2 \in [0, 1)$ satisfy $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$. Let $\alpha_t = \frac{\alpha}{\sqrt{t}}$ and $\beta_{1,t} = \beta_1 \lambda^{t-1}$, $\lambda \in (0, 1)$. Adam achieves the following guarantee, for all $T \geq 1$.

$$R(T) \leq \frac{D^2}{2\alpha(1-\beta_1)} \sum_{i=1}^d \sqrt{T\widehat{v}_{T,i}} + \frac{\alpha(\beta_1+1)G_\infty}{(1-\beta_1)\sqrt{1-\beta_2}(1-\gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \sum_{i=1}^d \frac{D_\infty^2 G_\infty \sqrt{1-\beta_2}}{2\alpha(1-\beta_1)(1-\lambda)^2}$$

Proof. Using Lemma 10.2, we have,

$$f_t(\theta_t) - f_t(\theta^*) \leq g_t^T(\theta_t - \theta^*) = \sum_{i=1}^d g_{t,i}(\theta_{t,i} - \theta_{*,i}^*)$$

From the update rules presented in algorithm 1,

$$\begin{aligned} \theta_{t+1} &= \theta_t - \alpha_t \widehat{m}_t / \sqrt{\widehat{v}_t} \\ &= \theta_t - \frac{\alpha_t}{1 - \beta_1^t} \left(\frac{\beta_{1,t}}{\sqrt{\widehat{v}_t}} m_{t-1} + \frac{(1 - \beta_{1,t})}{\sqrt{\widehat{v}_t}} g_t \right) \end{aligned}$$

We focus on the i^{th} dimension of the parameter vector $\theta_t \in R^d$. Subtract the scalar $\theta_{*,i}^*$ and square both sides of the above update rule, we have,

$$(\theta_{t+1,i} - \theta_{*,i}^*)^2 = (\theta_{t,i} - \theta_{*,i}^*)^2 - \frac{2\alpha_t}{1 - \beta_1^t} \left(\frac{\beta_{1,t}}{\sqrt{\widehat{v}_{t,i}}} m_{t-1,i} + (1 - \frac{\beta_{1,t}}{\sqrt{\widehat{v}_{t,i}}} g_{t,i}) \right) (\theta_{t,i} - \theta_{*,i}^*) + \alpha_t^2 \left(\frac{\widehat{m}_{t,i}}{\sqrt{\widehat{v}_{t,i}}} \right)^2$$

We can rearrange the above equation and use Young's inequality, $ab \leq a^2/2 + b^2/2$. Also, it can be shown that $\sqrt{\widehat{v}_{t,i}} = \sqrt{\sum_{j=1}^t (1 - \beta_2) \beta_2^{t-j} g_{j,i}^2} / \sqrt{1 - \beta_2^t} \leq \|g_{1:t,i}\|_2$ and $\beta_{1,t} \leq \beta_1$. Then

$$\begin{aligned} g_{t,i}(\theta_{t,i} - \theta_{*,i}^*) &= \frac{(1 - \beta_1^t) \sqrt{\widehat{v}_{t,i}}}{2\alpha_t(1 - \beta_{1,t})} \left((\theta_{t,i} - \theta_{*,i}^*)^2 - (\theta_{t+1,i} - \theta_{*,i}^*)^2 \right) \\ &\quad + \frac{\beta_{1,t}}{(1 - \beta_{1,t})} \frac{\widehat{v}_{t-1,i}^{\frac{1}{4}}}{\sqrt{\alpha_{t-1}}} (\theta_{*,i}^* - \theta_{t,i}) \sqrt{\alpha_{t-1}} \frac{m_{t-1,i}}{\widehat{v}_{t-1,i}^{\frac{1}{4}}} + \frac{\alpha_t(1 - \beta_1^t) \sqrt{\widehat{v}_{t,i}}}{2(1 - \beta_{1,t})} \left(\frac{\widehat{m}_{t,i}}{\sqrt{\widehat{v}_{t,i}}} \right)^2 \\ &\leq \frac{1}{2\alpha_t(1 - \beta_1)} \left((\theta_{t,i} - \theta_{*,i}^*)^2 - (\theta_{t+1,i} - \theta_{*,i}^*)^2 \right) \sqrt{\widehat{v}_{t,i}} + \frac{\beta_{1,t}}{2\alpha_{t-1}(1 - \beta_{1,t})} (\theta_{*,i}^* - \theta_{t,i})^2 \sqrt{\widehat{v}_{t-1,i}} \\ &\quad + \frac{\beta_1 \alpha_{t-1}}{2(1 - \beta_1)} \frac{m_{t-1,i}^2}{\sqrt{\widehat{v}_{t-1,i}}} + \frac{\alpha_t}{2(1 - \beta_1)} \frac{\widehat{m}_{t,i}^2}{\sqrt{\widehat{v}_{t,i}}} \end{aligned}$$

We apply Lemma 10.4 to the above inequality and derive the regret bound by summing across all the dimensions for $i \in 1, \dots, d$ in the upper bound of $f_t(\theta_t) - f_t(\theta^*)$ and the sequence of convex functions for $t \in 1, \dots, T$:

$$\begin{aligned} R(T) &\leq \sum_{i=1}^d \frac{1}{2\alpha_1(1 - \beta_1)} (\theta_{1,i} - \theta_{*,i}^*)^2 \sqrt{\widehat{v}_{1,i}} + \sum_{i=1}^d \sum_{t=2}^T \frac{1}{2(1 - \beta_1)} (\theta_{t,i} - \theta_{*,i}^*)^2 \left(\frac{\sqrt{\widehat{v}_{t,i}}}{\alpha_t} - \frac{\sqrt{\widehat{v}_{t-1,i}}}{\alpha_{t-1}} \right) \\ &\quad + \frac{\beta_1 \alpha G_\infty}{(1 - \beta_1) \sqrt{1 - \beta_2} (1 - \gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 + \frac{\alpha G_\infty}{(1 - \beta_1) \sqrt{1 - \beta_2} (1 - \gamma)^2} \sum_{i=1}^d \|g_{1:T,i}\|_2 \\ &\quad + \sum_{i=1}^d \sum_{t=1}^T \frac{\beta_{1,t}}{2\alpha_t(1 - \beta_{1,t})} (\theta_{*,i}^* - \theta_{t,i})^2 \sqrt{\widehat{v}_{t,i}} \end{aligned}$$